

LIBBY CASE REPORT

# demo-nsc lc-egfr-l858r-post-osi-d3m0

*Multi-agent decision support — clinician track*

Generated 2026-05-07

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*Libby is an experimental decision-support tool. The recommendations in this report have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.*

# Executive summary

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**Case:** demo-nsclc-egfr-l858r-post-osi-d3m0 **Question:** EGFR L858R-mutant stage IV lung adenocarcinoma with on-target MET amplification (GCN 8.2) after 22 months of first-line osimertinib. T790M is absent. What's the next line, given a working artist who needs her hands and prefers trials and oral therapy? **Evidence base:** 4 trials, 5 clinical-evidence rows, 4 preclinical rows. Three ranked interventions. Board agreement scores span +0.8 to -0.4, so rank 1 carries near-consensus and rank 2 carries an active veto.

## What this report covers

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The PI's synthesis of the post-osimertinib decision when MET amplification is the dominant resistance driver. There are no scenario branches in this case; the three ranked options are unconditional, and the gap between rank 1 and rank 2 is large enough that the report reads as one main recommendation plus two transparency entries.

## Top-line findings

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- MET amplification at GCN 8.2 by FISH, with T790M absent, points cleanly at a MET-directed combination. Savolitinib plus osimertinib is the option that fits both the biology and the patient's stated preferences.
- The board endorsed rank 1 four-to-one. The consensusite flagged that savo-osi is not yet inside NCCN; the resolution is to enroll on SAFFRON or SACHI rather than treat off-trial, which the patient already prefers.
- The conservative vetoed rank 2 (amivantamab + lazertinib) on severe-neuropathy and IV-burden grounds. Critic and advocate also dissented. Two personas endorsed it on guideline alignment and effect size, so it stays on the table as *considered\_with\_caveats* rather than being dropped — but override of the veto is not recommended unless rank 1 fails.
- Rank 3 (patritumab deruxtecan) triggers the patient's alopecia veto outright. Only the risktaker advanced it. It's listed for transparency, not as a live option.

## Recommendation summary

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1. **Savolitinib + osimertinib (preferably on SAFFRON / SACHI)** — recommended. ORR ~30% in the MET-amp post-EGFR-TKI stratum (TATTON expansion, PMID 32679432). All-oral; clears every stated veto; trial-route resolves the guideline gap.
2. **Amivantamab + lazertinib** — considered with caveats. ORR 36% in the CHRYSALIS-2 post-osimertinib subset. Conservative veto holds on neuropathy and IV chair time, both of which matter unusually for a patient whose work depends on her hands.
3. **Patritumab deruxtecan (HER3-DXd)** — considered with caveats. Triggers the alopecia veto. Surfaced so the user can revisit it with her oncologist if she chooses to re-weight that veto.

## What this report does *\*not\** cover

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Dose adjustments, monitoring schedules, sequencing across later lines, payer or access logistics, or institution-specific trial slot availability. Those belong to the treating team.

## How to use this report

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Each ranked option carries the board's agreement state (endorsed, dissent, veto by persona) and one to three anchor citations. The full per-trial extraction lives on the live case page; this PDF is the synthesis.

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<!-- libby:downloads:begin -->

## Downloads

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- [Clinician PDF report](#) — ranked recommendations + evidence + sources
- [Patient/caregiver PDF](#) — plain-language summary
- [Target validation paths](#) — diagnostic + biomarker workup that hardens the targetable-feature call
- [Access guide \(web\)](#) — same access guide in an in-browser sortable table
- [Master manuscripts table \(PDF\)](#) — every paper considered — n, effect, variance, toxicities
- [Master manuscripts table \(web\)](#) — same inventory in a sortable in-browser table
- [Self-contained HTML](#) — recommendations table that opens offline

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## Research question

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A working artist in her 60s has stage IV EGFR L858R lung adenocarcinoma. She got 22 months out of first-line osimertinib (PR), and now she is progressing. The tumor is T790M-negative but MET-amplified at GCN 8.2 by FISH. What should the next line of therapy target, given that she needs her hands, prefers oral therapy, and is open to trials?

## Patient profile (scrubbed)

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- **Primary site / histology:** lung adenocarcinoma
- **Stage:** IV
- **Performance status:** ECOG 1
- **Age band:** 60-69
- **Biomarkers:** EGFR L858R (present); EGFR T790M (absent); MET amplification (GCN 8.2 by FISH); PD-L1 TPS 10%
- **Prior therapy:** 1L osimertinib, PR x 22 months — now progressed
- **Targetable features:** EGFR L858R (sensitizing); MET amplification (likely on-target resistance)

## Preferences

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- **Efficacy/toxicity weight:** 0.55 (slight efficacy lean)
- **Toxicity vetoes:** severe neuropathy, cardiotoxicity, alopecia
- **Modality constraints:** oral preferred; no inpatient infusion; minimize IV chair time
- **Free text:** *"working artist — manual dexterity in hands matters more than typical"*
- **Trials preferred:** yes

<!-- libby:target-validation:begin -->

## Target validation paths

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EGFR L858R, MET amplification (GCN 8.2), and T790M-absent are all confirmed at decision-relevant resolution, so no diagnostic test gates the rank-1 trial. The validation work here is about resistance characterization at progression, not target confirmation. The goal is to make sure the case really is MET-amplified bypass resistance and not something more complicated wearing the same imaging signature.

## EGFR L858R

The most consequential miss right now would be SCLC histologic transformation, which is the highest-PPV resistance pattern that no liquid biopsy can confirm. Tumor NGS including TP53 and RB1 status is the screen: co-loss is the strongest predictor of transformation, with reported rates of 3–14% across post-osimertinib cohorts. If imaging morphology shifts or co-loss is detected, fresh re-biopsy with a neuroendocrine IHC panel (chromogranin, synaptophysin, INSM1, Ki-67) is the only way to confirm. ctDNA covering EGFR C797S and exon-20 inserts is the parallel on-target resistance check; a positive C797S result reframes the next-line conversation toward fourth-generation EGFR-TKIs or combination strategies.

## MET amplification

The FISH GCN 8.2 result clears SAFFRON's enrollment threshold ( $\geq 6$ ) cleanly, but MET amplification can be focal, and primary-vs-metastatic discordance is a documented confounder in this setting. A second-site FISH (or a matched archival block) refines confidence that the dominant disease, not just the biopsied site, is the amplified one. MET IHC (clone SP44) is the orthogonal-modality complement to the SAFFRON eligibility profile. A comprehensive NGS panel that picks up HER2 / BRAF / FGFR co-alterations frames durability expectations and the next-line plan.

## Where to order these assays

| Assay                                         | Provider                                         | Contact                    |
|-----------------------------------------------|--------------------------------------------------|----------------------------|
| ctDNA panel (EGFR C797S / T790M / exon-20)    | Guardant Health (Guardant360 CDx)                | test info · 1-855-698-8887 |
| ctDNA panel                                   | Foundation Medicine (FoundationOne Liquid CDx)   | test info · 1-888-988-3639 |
| ctDNA panel                                   | Tempus (xF+)                                     | test info · 1-800-739-4137 |
| ctDNA panel                                   | Caris Life Sciences (Caris Assure)               | test info · 1-888-979-8669 |
| ctDNA panel                                   | NeoGenomics Laboratories (NeoLAB Liquid)         | test info · 1-866-776-5907 |
| Tumor NGS for TP53 / RB1 + bypass alterations | Foundation Medicine (FoundationOne CDx)          | test info · 1-888-988-3639 |
| Tumor NGS                                     | Caris Life Sciences (Molecular Intelligence)     | test info · 1-888-979-8669 |
| Tumor NGS                                     | Tempus (xT)                                      | test info · 1-800-739-4137 |
| Tumor NGS                                     | NeoGenomics Laboratories (NeoTYPE Comprehensive) | test info · 1-866-776-5907 |
| Tumor NGS                                     | Memorial Sloan Kettering (MSK-IMPACT)            | test info · 1-212-639-2000 |
| MET FISH (gene/CEP7 ratio)                    | NeoGenomics Laboratories                         | test info · 1-866-776-5907 |
| MET FISH                                      | Mayo Clinic Laboratories                         | test info · 1-800-533-1710 |

|                                      |                             |                            |
|--------------------------------------|-----------------------------|----------------------------|
| MET FISH                             | LabCorp / Esoterix Oncology | test info · 1-800-345-4363 |
| MET IHC (clone SP44)                 | Mayo Clinic Laboratories    | test info · 1-800-533-1710 |
| MET IHC (clone SP44)                 | ARUP Laboratories           | test info · 1-800-242-2787 |
| Neuroendocrine IHC panel (re-biopsy) | Mayo Clinic Laboratories    | test info · 1-800-533-1710 |
| Neuroendocrine IHC panel             | ARUP Laboratories           | test info · 1-800-242-2787 |

*Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.*

<!-- libby:target-validation:end -->

## Scope summary

The dossier behind this page contains 4 trials, 5 clinical-evidence rows, and 4 preclinical rows. Three interventions get ranked. There is no scenario branching because every biomarker on this patient is **confirmed**. Board agreement scores run from +0.8 at rank 1 down to -0.4 at rank 2, where there is an active veto, so the gap between the top two picks is wide and does real work in the ranking.

## Cross-cutting caveat (read first)

MET amplification at GCN 8.2 with T790M absent points cleanly at MET-directed combination therapy. The patient's modality and toxicity preferences then close off the most-cited alternative. That is the story behind every rank below.

- T790M-negative status takes second-generation osimertinib re-challenge off the table as a sequencing move. The resistance is not gatekeeper-mutation-driven.
- MET amplification accounts for roughly 15–30% of post-osimertinib resistance at this GCN range. Savolitinib + osimertinib is the mechanism-matched combination with replicated evidence in this exact subset.
- The patient's manual-dexterity-driven veto on **severe neuropathy** plus **minimize IV chair time** rules out amivantamab + lazertinib at rank 1 even though the regimen carries NCCN cat-2A status. The conservative's toxicity veto on this rec is the load-bearing dissent.
- The **alopecia veto** alone forecloses patritumab deruxtecan independent of efficacy. The dossier keeps it on the page for transparency, not as a live option.

## Intervention grouping

Three buckets, sorted by how cleanly each one matches the resistance biology.

- MET-targeted EGFR-TKI combinations (mechanism-matched, all-oral): savolitinib + osimertinib via the SAFFRON / SACHI phase-3 program. The mechanistic anchor is the TATTON expansion ([PMID 32679432](#)).
- EGFR + MET bispecific paired with a 3G TKI (broader-spectrum, IV-anchored): amivantamab + lazertinib via CHRYSALIS-2 ([PMID 36720074](#)).
- HER3-directed antibody-drug conjugate (mechanism-agnostic post-TKI option): patritumab deruxtecan via

## Top interventions

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### Rank 1. Savolitinib + osimertinib (preferably on SAFFRON / SACHI)

*Mechanism-matched, all-oral, clears every preference axis. Four of five personas endorsed. Consensusite's qualified guideline-fit concern resolves on trial enrollment.*

#### Evidence base

The mechanistic and clinical anchor is the TATTON expansion (PMID 32679432; n=69; ORR 30%, 95% CI 19–42 in the EGFR-mut + MET-amp post-EGFR-TKI subset). The phase-3 confirmatory study (NCT05261399; SAFFRON; n=324; savo + osi vs platinum + pemetrexed; PFS endpoint) is recruiting and reads out into the same biomarker-defined population. The patient's MET-amp GCN 8.2 by FISH clears the GCN  $\geq 6$  / IHC 3+ enrollment threshold without needing a borderline call.

#### Likelihood of desired effect

Mechanism concordance is high. MET amplification IS the on-target resistance mechanism this combination targets, and the patient's biomarker fits the studied stratum exactly. The TATTON ORR (~30%) is real, replicated within the program, and pre-specified to this subset. The phase-3 readout will quantify durability and PFS gain over chemotherapy.

#### Toxicity profile

- LFT elevation, manageable with dose reduction
- Peripheral edema
- Fatigue
- All-oral, with no inpatient component
- No severe-neuropathy signal in either TATTON or the broader SAFFRON program

User vetoes (severe neuropathy, cardiotoxicity, alopecia) all clear. Modality constraints (oral, no inpatient, minimize IV) all clear.

#### Counter-productive mechanisms / dissent

Endorsed by risktaker, conservative, critic, and advocate. Consensusite raised a *qualified guideline-fit* concern: the combination is not currently NCCN-listed as a recommended regimen, so off-trial use is off-label. The resolution is trial enrollment, which the patient already prefers. No persona dissented or vetoed.

#### Practical considerations

- All-oral: savolitinib 300 mg PO BID + osimertinib 80 mg PO daily
- Trial enrollment is the guideline-aligned route
- Off-trial use is off-label, so flag it with the treating team
- LFT monitoring at baseline plus every 2–4 weeks early in therapy

#### Why this rank

Strong combined evidence-and-preference fit. Every preference axis clears, the mechanism matches the

biomarker, and the agreement\_score (+0.8) is the highest in the dossier. The rank-1/rank-2 gap is +1.2 (vs amivantamab + lazertinib at -0.4), driven both by the active veto on rank 2 and by the better preference fit here.

Per-trial detail

| Therapeutic agent                                                                                    | Efficacy                      | Toxicity                                                  | Reference     |
|------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------|---------------|
| Savolitinib + osimertinib — TATTON expansion (single-arm phase 1b, n=69 EGFR-mut + MET-amp post-TKI) | ORR 30% (95% CI 19–42)        | LFT elevation, peripheral edema, fatigue (all manageable) | PMID 32679432 |
| Savolitinib + osimertinib — SAFFRON (phase 3 RCT, recruiting, n=324)                                 | PFS endpoint; outcome pending | Per TATTON safety profile + comparator chemo arm          | NCT05261399   |

Rank 2. Amivantamab + lazertinib

\*Real efficacy signal, NCCN cat-2A, but **conservative vetoed on neuropathy + IV-burden grounds**. Considered with caveats; override not recommended absent rank-1 failure.\*

Evidence base

CHRYSLIS-2 (PMID 36720074; single-arm phase 1/2; n=162; post-osimertinib EGFR-mutant biomarker subset including MET-amp): ORR 36% (95% CI 28–45). NCCN NSCLC v3.2025 category 2A. ESMO MCBS B.

Likelihood of desired effect

Effect size is real and slightly larger than rank 1's ORR. Biomarker fit is acceptable (EGFR L858R + MET-amp permitted), but the trial is not enriched specifically for MET-amplified post-osimertinib progressors at this GCN range, so the subset estimate is less granular than TATTON's.

Toxicity profile

- **Severe neuropathy** in combination data, which triggers the user veto
- IRR grade 3+ ~7%
- Rash
- Inpatient or extended chair-time IV cycle 1, which conflicts with both modality preferences (no inpatient infusion; minimize IV chair time)

For a working artist whose manual dexterity matters more than typical, the neuropathy risk is decision-relevant beyond the categorical veto.

Counter-productive mechanisms / dissent

**Conservative issued a toxicity veto on stated severe-neuropathy and IV-burden grounds.** Advocate dissented on preference fit. Critic dissented on evidence-quality grounds (single-arm phase 1/2, no randomized comparator in the post-osimertinib subset). Risktaker endorsed on effect size; consensusite endorsed on NCCN guideline alignment.

The veto is honored, not silently dropped. Per PI synthesis rules this row stays on the page as

considered\_with\_caveats so the user sees what was considered. Override of the conservative's veto is **not** recommended absent failure of the rank-1 option.

## Practical considerations

- Amivantamab 1050 mg IV weekly cycle 1 then q2w + lazertinib 240 mg PO daily
- Subcutaneous amivantamab is in development; AE re-profiling pending
- NCCN-recommended; off-label considerations are weaker than for rank 1

## Why this rank

Below savolitinib + osimertinib because of the active toxicity veto plus the modality conflict, even though the ORR is slightly larger. Above patritumab deruxtecan because two personas affirmatively endorsed and the alopecia veto on rank 3 is unconditional.

## Per-trial detail

| Therapeutic agent                                                                    | Efficacy               | Toxicity                                           | Reference                     |
|--------------------------------------------------------------------------------------|------------------------|----------------------------------------------------|-------------------------------|
| Amivantamab + lazertinib — CHRYSLIS-2 (single-arm phase 1/2, n=162 post-osimertinib) | ORR 36% (95% CI 28–45) | Severe neuropathy (combination); IRR G3+ ~7%; rash | <a href="#">PMID 36720074</a> |

## Rank 3. Patritumab deruxtecan (HER3-DXd)

\*Real efficacy, but **the alopecia veto forecloses it**. Surfaced only so the user can re-weight the veto with her oncologist if she chooses.\*

## Evidence base

HERTHENA-Lung01 ([PMID 37563559](#); single-arm phase 2; n=225; EGFR-mutant NSCLC after EGFR-TKI): ORR 29.8% (95% CI 23.9–36.2). Effect size is comparable to ranks 1 and 2.

## Likelihood of desired effect

The mechanism (HER3-targeted antibody-drug conjugate) is mechanism-agnostic relative to the patient's MET-amp resistance. It targets HER3 expression rather than the resistance pathway. The effect is real, but the mechanism fit is weaker than at rank 1.

## Toxicity profile

- ILD grade 3+ ~5%
- **Alopecia**, which triggers the user veto
- Thrombocytopenia
- IV q3w

## Counter-productive mechanisms / dissent

Only the risktaker advanced this rec. Advocate did not advance it because the alopecia veto was already on the table. No formal veto was issued because the dossier surfaced the rec only after ranks 1 and 2 were established.

### Practical considerations

- 5.6 mg/kg IV q3w
- Not currently NCCN-endorsed as standard for this setting
- ILD risk requires baseline pulmonary function plus ongoing monitoring

### Why this rank

Below ranks 1 and 2 because the alopecia veto is unconditional. The rec is on the page for transparency: the user can ask her oncologist whether re-weighting the alopecia veto changes the calculus. That is a conversation, not a recommendation.

### Per-trial detail

| Therapeutic agent                                                                       | Efficacy                     | Toxicity                                   | Reference                     |
|-----------------------------------------------------------------------------------------|------------------------------|--------------------------------------------|-------------------------------|
| Patritumab deruxtecan —<br>HERTHENA-Lung01 (single-arm<br>phase 2, n=225 post-EGFR-TKI) | ORR 29.8% (95% CI 23.9–36.2) | ILD G3+ ~5%; alopecia;<br>thrombocytopenia | <a href="#">PMID 37563559</a> |

### Classes examined but not ranked

- Osimertinib re-challenge or dose escalation: T790M is absent, so the gatekeeper-mutation pathway is not the resistance mechanism, and re-challenge has no biomarker rationale.
- MET-targeted monotherapy (capmatinib, tepotinib): the mechanism overlaps with the rank-1 savolitinib + osimertinib combo, but the approved indications cover METex14 rather than amplification, so the biomarker subset does not match.

### Ranked prioritization

| Rank | Status | Intervention                                                          | Endorsed by | Dissent | Veto | Likelihood                                          | Toxicity burden        | Why this rank                            |
|------|--------|-----------------------------------------------------------------------|-------------|---------|------|-----------------------------------------------------|------------------------|------------------------------------------|
| 1    |        | Savolitinib +<br>osimertinib<br>(preferably on<br>SAFFRON /<br>SACHI) | <span       | —       | —    | High<br>mechanism<br>fit; ORR<br>~30%<br>replicated | LFT, edema,<br>fatigue | Mechanism +<br>preferences<br>both clean |
|      |        |                                                                       | <span       |         |      |                                                     |                        |                                          |
|      |        |                                                                       | <span       |         |      |                                                     |                        |                                          |

|   |                                  |       |       |       |                    |                              |                                 |
|---|----------------------------------|-------|-------|-------|--------------------|------------------------------|---------------------------------|
| 2 | Amivantamab + lazertinib         | <span | <span | <span | ORR 36% in subset  | Severe neuropathy, IV burden | Active veto + modality conflict |
|   |                                  | <span | <span |       |                    |                              |                                 |
| 3 | Patritumab deruxtecan (HER3-DXd) | <span | —     | —     | ORR 29.8% post-TKI | ILD ~5%; alopecia            | Triggers alopecia veto outright |

# Caveats

- The evidence base is modest in size. Rank 1's anchor is a 69-patient TATTON expansion plus a recruiting phase 3. Rank 2's anchor is a 162-patient single-arm. Rank 3's anchor is a 225-patient single-arm. None has a randomized comparator in this patient's exact biomarker stratum yet.
- No OS data exist for any of the three ranked interventions in the post-osimertinib MET-amp setting specifically. Effect estimates are ORR-based, and durability is the open question for all three.
- Two of the three rec-ranks turn on user vetoes (neuropathy, alopecia) tied to the working-artist context. The board treated those as load-bearing, not disposable.
- What would change the ranking:
  - A SAFFRON OS readout that confirms a meaningful PFS / OS benefit would tighten rank 1's confidence and could push it from "preferred" to "guideline-fit."
  - A MET-amp-stratified amivantamab + lazertinib randomized readout that mitigates neuropathy (for example with the SC formulation) could narrow the rank-1 / rank-2 gap, but the modality preference would still anchor rank 1 above it.
  - If the user re-weighted the alopecia veto, patritumab deruxtecan would move from rank 3 ( `considered_with_caveats`) to rank 2.
  - Failure of the rank-1 option (intolerance, progression, slot unavailability) is the explicit precondition for revisiting rank 2 with the conservative's veto on the table.
- Re-scoping caveat. If the patient's clinical state changes (CNS progression, declining ECOG) or if the modality preference relaxes, the ranking shifts. The current page assumes the stated profile and preferences hold.

# Sources

## PubMed (PMID):

- [32679432](#) — Sequist et al., TATTON expansion, *Lancet Oncol* 2020
- [36720074](#) — Cho et al., CHRYSALIS-2, *JCO* 2023
- [37563559](#) — Yu et al., HERTHENA-Lung01, *JCO* 2023

## ClinicalTrials.gov (NCT):

- [NCT05261399](#) — SAFFRON (savolitinib + osimertinib phase 3)
- [NCT04077463](#) — CHRYSALIS-2 (amivantamab + lazertinib)
- [NCT04619004](#) — HERTHENA-Lung01 (patritumab deruxtecan)
- [NCT02143466](#) — TATTON

## Transparency artifacts

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- [Trial table](#) — 4 trials in the dossier
- [Evidence list](#) — 5 clinical-evidence rows + 4 preclinical rows
- [Tumor-board transcript](#) — 5 positions, 20 cross-critiques
- [Recommendations table](#) — full ranked detail
- [Plain-language summary](#) — patient/caregiver track

## Run log

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This is a demo case authored in May 2026 with synthetic data so the pipeline can be smoke-tested. No real patient is represented. The page was re-rendered into the shieldbreak-flavored layout the same month after Libby's PI authoring contract was updated to mirror [pirl-unc/io-shieldbreak](#), then re-run with the humanizer pass once that was added to the contract.

## Sources

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### PubMed (PMID):

- [32679432](#)
- [36720074](#)
- [37563559](#)

### ClinicalTrials.gov (NCT):

- [NCT05261399](#)